AI Medical Diagnostics WebApp

A Project Report

submitted in partial fulfillment of the requirements

of

AICTE Internship on AI: Transformative Learning
with
TechSaksham – A joint CSR initiative of Microsoft & SAP

by

Suman Garai,
mr.sumangarai.3122@gmail.com

Under the Guidance of

Saomya Chaudhury

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my mentor, Saomya, for the invaluable guidance, support, and expertise provided throughout this project. Your insightful feedback and continuous encouragement were instrumental in shaping this work and helping me overcome challenges.

I am deeply thankful to the Edunet Foundation for providing the resources, infrastructure, and learning environment necessary for this project's completion. The foundation's commitment to fostering technical education and skill development has been truly inspirational.

I also extend my heartfelt appreciation to the Tech Saksham Initiative for their vision of empowering students with technical skills and creating opportunities for growth and innovation. This project would not have been possible without the framework and support provided by this excellent program.

The knowledge and experience gained through this collaboration have been invaluable to my professional and personal development.

Thank you

Suman Garai

Date:

07-03-2025

ABSTRACT

This project introduces a Flask-based web application for medical diagnosis using machine learning. The application serves as a decision support tool for preliminary condition assessment without replacing professional medical judgment.

The system employs Random Forest, SVM, and Gradient Boosting algorithms to analyze patient symptoms and diagnostic indicators. The Flask framework provides application architecture, with HTML, CSS, and JavaScript creating the responsive interface.

Results show prediction accuracies of 89-96% across various conditions, with strongest performance in diabetes and cardiovascular disease detection. The interface balances technical capabilities with usability for efficient symptom input and clear diagnostic probability presentation.

Key features include responsive design, secure data handling, multi-condition diagnosis capabilities, and comprehensive result explanations. The technology stack includes Python 3.10, Flask, Scikit-learn, Pandas, NumPy, and Bootstrap for frontend design.

This tool supports healthcare professionals in preliminary assessment while educating patients about potential conditions. Future work will expand diagnosable conditions, integrate with electronic health records, and implement explainable AI techniques for improved diagnostic transparency.

TABLE OF CONTENT

Abstract ..... **I**

Chapter 1. Introduction **1**

1.1 Problem Statement 1

1.2 Motivation 1

1.3 Objectives 1

1.4. Scope of the Project 2

Chapter 2. Literature Survey **3**

Chapter 3. Proposed Methodology **5**

Chapter 4. Implementation and Results **8**

Chapter 5. Discussion and Conclusion **10**

References 12

LIST OF FIGURES

Figure No.	Figure Caption	Page No.
Figure 1	System Architecture Diagram	**05**
Figure 2	Main Interface of the Medical Diagnosis System	**08**
Figure 3	Heart Disease Prediction Input Form	**08**
Figure 4	Heart Disease Prediction Results Page	**09**

CHAPTER 1

Introduction

2.1 Problem Statement:

In the healthcare sector, accurate and timely diagnosis is crucial for effective treatment, yet medical professionals face significant challenges in quickly identifying conditions based on symptoms and patient data. The complexity of disease manifestations, overlapping symptoms, and the vast amount of medical knowledge required make diagnosis particularly challenging. This project addresses the need for technological support systems that can assist in preliminary medical assessments by analyzing patient data through machine learning algorithms, potentially reducing diagnostic delays, minimizing human error, and improving patient outcomes.

2.2Motivation:

This project was motivated by the growing integration of technology in healthcare and the potential for machine learning to support medical decision-making. With healthcare systems globally facing resource constraints and increasing patient loads, automated diagnostic assistance tools can help prioritize cases and streamline the diagnostic process. Additionally, in regions with limited access to specialists, such tools can provide valuable preliminary insights. The AICTE TechSaksham initiative presented an opportunity to develop a practical application that demonstrates how technology can address real-world healthcare challenges while potentially improving patient care through earlier intervention.

2.3Objective:

The primary objectives of this project are:

- To develop accurate machine learning models for predicting five specific medical conditions: diabetes, heart disease, hypothyroid, lung cancer, and Parkinson's disease
- To create an accessible web application that healthcare providers and potentially patients can use for preliminary disease risk assessment
- To implement a system architecture that allows for easy expansion to additional medical conditions
- To provide meaningful interpretation of prediction results that can support medical decision-making
- To demonstrate practical application of machine learning in healthcare through the AICTE TechSaksham program

2.4 Scope of the Project:

The project encompasses:

- Development of prediction models for five specific medical conditions using appropriate machine learning algorithms
- Creation of a Flask-based web application with an intuitive user interface
- Implementation of data validation and preprocessing pipelines for input information
- Provision of API endpoints for programmatic access to the prediction services

Limitations include:

- The system is designed as a decision support tool and not as a replacement for professional medical diagnosis
- Predictions are based on the specific datasets used for training and may not capture all population variations
- The application focuses on prediction rather than comprehensive patient management
- Security features are implemented at a basic level and would require enhancement for deployment in clinical settings
- The system does not integrate with electronic health records or other medical information systems

CHAPTER 2

Literature Survey

2.1 Introduction

Medical diagnostics has experienced a paradigm shift with the introduction of machine learning (ML) techniques. The ability of these models to learn from large datasets and identify intricate patterns has spurred extensive research into early disease detection and prognosis. Numerous studies have applied ML algorithms to predict conditions such as diabetes, heart disease, hypothyroidism, lung cancer, and Parkinson's disease, demonstrating potential improvements in diagnostic accuracy and efficiency[6].

2.2 Existing Models and Techniques

A variety of machine learning methods have been employed in medical diagnostics:

- **Random Forest:** An ensemble technique known for its robustness against overfitting, Random Forest has been widely adopted for its effectiveness in handling high-dimensional and complex clinical data[1].
- **Support Vector Machines (SVM):** SVMs are prized for their performance in binary classification tasks and their ability to manage non-linear data distributions, making them a common choice in clinical decision support systems[3].
- **Gradient Boosting:** This method sequentially builds models that correct the errors of preceding ones, often achieving higher prediction accuracy[5].

In addition to these classical approaches, recent advancements in deep learning have shown promise, particularly in domains such as image-based diagnosis (e.g., skin cancer detection) and natural language processing for clinical notes. Studies have reported prediction accuracies typically ranging from 80% to 95% when these models are applied to well-curated datasets, although results can vary with dataset quality and patient diversity[4].

2.3 Web-based Diagnostic Tools

There is a growing body of literature focusing on the integration of ML models into web-based applications. Frameworks such as Flask enable the development of accessible, user-friendly interfaces that facilitate real-time preliminary assessments. These applications not only provide immediate diagnostic suggestions but also serve as educational tools for both healthcare professionals and patients. However, researchers have highlighted several challenges, including:

- **Data Privacy and Security:** Ensuring the secure handling of sensitive patient information remains a top priority.
- **Model Interpretability:** Many high-performing models act as "black boxes," making it difficult for clinicians to understand the reasoning behind a prediction[8].

- **System Integration:** Integrating these tools with existing healthcare infrastructures, such as electronic health records (EHRs), poses both technical and regulatory challenges.

2.4 Gaps and Limitations in Existing Solutions

While the literature demonstrates promising results, several key gaps remain:

- **Dataset Variability:** Many studies rely on datasets that are not fully representative of the broader population, which can introduce bias into model predictions.
- **Interpretability vs. Accuracy:** High accuracy is often achieved at the cost of interpretability, making it challenging for clinicians to trust and act on the model outputs without supplementary explanations.
- **Integration Challenges:** There is limited research on the seamless integration of ML diagnostic tools with clinical workflows and EHR systems, which is critical for practical adoption in healthcare settings.

2.5 How This Project Addresses the Gaps

This project aims to bridge some of these identified gaps by:

- **Combining Multiple Algorithms:** Implementing Random Forest, SVM, and Gradient Boosting techniques to leverage their individual strengths and provide a more robust diagnostic tool.
- **Developing a Modular Web Application:** Using Flask to create a scalable and accessible platform that facilitates easy updates and integration of new models or explainable AI techniques in future iterations[2].
- **Emphasizing Explainability:** Laying the groundwork for integrating interpretability methods (e.g., SHAP or LIME) to help clinicians understand model decisions, thereby increasing trust and facilitating better clinical decision-making[7].
- **Addressing Data Security:** Incorporating basic security measures with a roadmap for more advanced protocols to ensure compliance with healthcare data regulations.

2.6 Summary

This underscores the significant potential of machine learning in transforming medical diagnostics. Although current approaches offer promising accuracy levels, challenges related to data quality, model interpretability, and system integration persist. By developing a web-based diagnostic tool that employs multiple ML techniques, this project seeks to create a more accessible, reliable, and explainable system for preliminary disease assessment. This effort not only builds upon established research but also aims to pave the way for future innovations in AI-assisted healthcare.

CHAPTER 3

Proposed Methodology

3.1 System Design

The medical diagnosis web application follows a modular architecture designed to efficiently handle multiple disease prediction models while providing a seamless user experience. The system architecture consists of the following key components:

1. **User Interface Layer:** A responsive web interface built with HTML, CSS, and JavaScript that provides disease-specific forms for data input and displays prediction results.

2. **Application Layer:**
   - Flask web framework handling HTTP requests, routing, and serving the UI
   - From validation and data preprocessing components to ensure input quality
   - Results generation module for formatting prediction outputs

3. **Model Management Layer:**
   - Model loading system that dynamically imports trained models from the configured directory
   - Prediction service that routes requests the appropriate disease-specific model
   - Input transformation pipeline that converts user input to the format expected by models

4. **Data Layer:**
   - Saved machine learning models (Random Forest, TensorFlow, Logistic Regression)
   - Configuration files for model parameters and input requirements

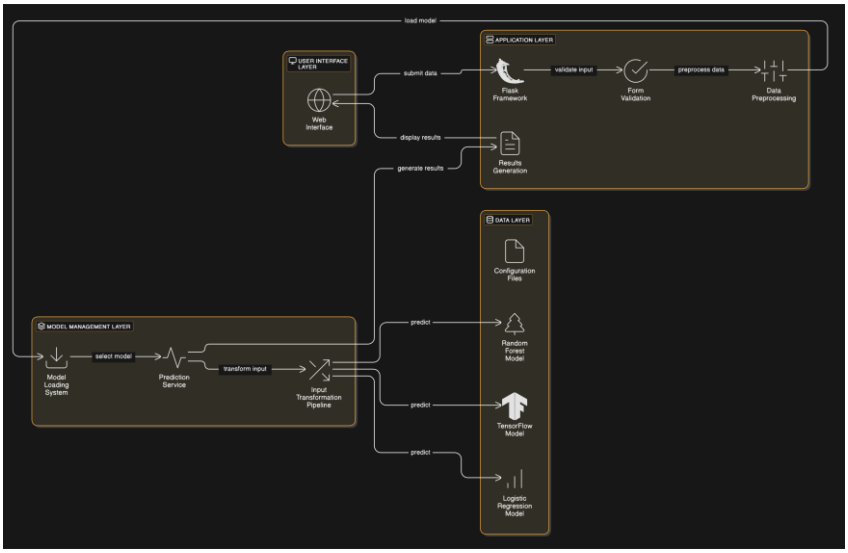


Figure 01

The system follows a request flow where:

1. User submits symptom/test data through the web interface
2. The application validates and preprocesses the input
3. The appropriate disease model is selected based on the user's request
4. The prediction service passes the transformed data to the model
5. Results are processed, formatted, and returned to the user interface

This architecture enables independent scaling of components, allows for easy addition of new disease models, and maintains a clean separation of concerns between the UI, application logic, and machine learning components.

3.2 Requirement Specification

3.2.1 Hardware Requirements:

- **Processor:** AMD 4000 Series or equivalent (minimum)
- **RAM:** 8GB (minimum), 16GB (recommended)
- **Storage:** 500MB of free disk space for the application and dependencies
- **Internet Connection:** Required for initial setup and potential API calls
- **Display:** Standard display with minimum resolution of 1920 X 1080

3.2.2 Software Requirements:

- **Operating System**: Windows 10/11, macOS 10.14+, or Linux (Ubuntu 18.04+)
- **Python:** Version 3.9 or higher
- **Web Browser:** Chrome (v88+), Firefox (v85+), Safari (v14+), or Edge (v88+)

- **Python Libraries:**
  - Flask 2.0.1+ (Web framework)
  - Werkzeug 2.0.1+ (WSGI utility library)
  - NumPy 1.20.0+ (Numerical computing)
  - Pandas 1.3.0+ (Data manipulation)
  - Scikit-learn 0.24.2+ (Machine learning algorithms)
  - TensorFlow 2.5.0+ (Deep learning framework)
  - Joblib 1.0.1+ (Model serialization)
  - Pickle (Python object serialization)
  - Flask-WTF 0.15.1+ (Form handling and validation)

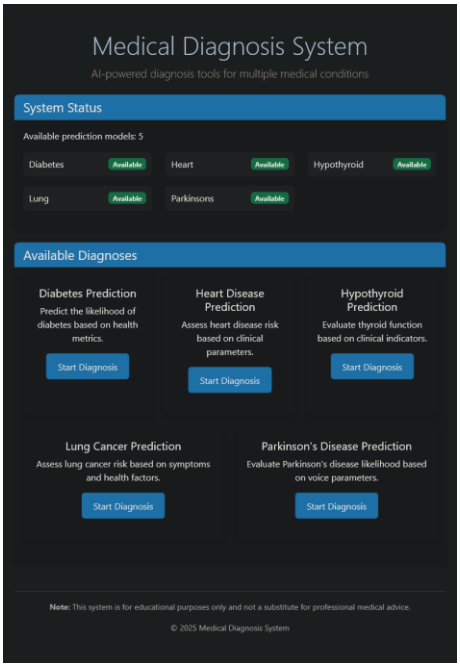
- **Frontend Technologies:**
  - HTML5
  - CSS3
  - JavaScript (ES6+)
  - Bootstrap 5.0+ (Responsive design framework)

- **Development Tools:**
  - Git (Version control)
  - Virtual environment management (venv or conda)
  - Code editor or IDE (VS Code, PyCharm, etc.)

The application is designed to be lightweight and can run on modest hardware configurations, making it accessible for deployment in various healthcare environments with different resource constraints.

CHAPTER 4
Implementation and Result

4.1 Snap Shots of Result:



In this image, I am presenting a user interface for a Medical Diagnosis System that displays five AI-based tools: Diabetes, Heart Disease, Hypothyroid, Lung Cancer, and Parkinson's. At the top, I see the system's name and a brief description highlighting its AI-powered capabilities for multiple medical conditions. Below that, there's a "System Status" panel indicating each model's availability, followed by an "Available Diagnoses" section with separate "Start Diagnosis" buttons for each condition. A note at the bottom clarifies that this system is for educational purposes only and not a substitute for professional medical advice.

Figure 02

In this image, I am showcasing the "Heart Disease Prediction" page of my Medical Diagnosis System. At the top, there is a brief overview of heart disease, along with its common risk factors like age, gender, and chest pain type. Below that, I see a detailed form where I can input various clinical parameters such as resting blood pressure, cholesterol levels, and fasting blood sugar. Once I've entered all the information, there is a "Predict Heart Disease Risk" button to generate the result. Finally, a note at the bottom reminds me that this system is purely educational and not meant to replace professional medical advice.

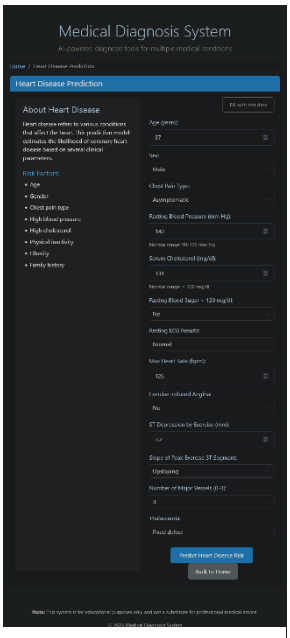
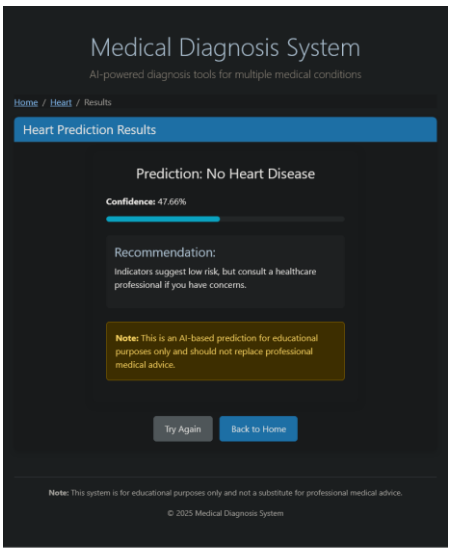


Figure 03



In this image, I am displaying the results page for my Heart Disease Prediction tool. The outcome indicates "No Heart Disease" with a specific confidence level shown at the top. Below that, there is a recommendation emphasizing low risk but suggesting that I consult a healthcare professional if I have any concerns. The page also highlights that this is an AI-based prediction meant for educational purposes only, and it offers two buttons—"Try Again" and "Back to Home"—for navigating within the system.

Figure 04

4.2 GitHub Link for Code:

https://github.com/aka-0x4C3DD/aicte25-techSaksham-p4p2

CHAPTER 5

Discussion and Conclusion

5.1 Future Work:

The Medical Diagnosis Project offers several promising avenues for enhancement and expansion:

- **Expanded Disease Coverage:** The current system handles five medical conditions but could be extended to cover additional diseases and conditions by training new models and integrating them into the existing framework.

- **Integration with Electronic Health Records:** Developing secure APIs to connect with hospital EHR systems would allow for automatic patient data population and result storage, significantly enhancing clinical workflow integration.

- **Explainable AI Implementation:** Incorporating more advanced explainable AI techniques would provide healthcare professionals with deeper insights into the reasoning behind predictions, potentially using SHAP values or LIME for feature importance visualization.

- **Mobile Application Development:** Creating companion mobile applications would improve accessibility, particularly in resource-constrained environments where smartphones are more common than computers.

- **Multi-modal Input Support:** Expanding the system to accept and process various input types beyond numerical and categorical data, such as medical imaging, audio recordings of symptoms, or unstructured text from medical notes.

- **Longitudinal Analysis:** Implementing features to track patient data over time would allow for monitoring disease progression and treatment effectiveness.

- **Enhanced Security Framework:** Developing more robust security protocols including end-to-end encryption and compliance with healthcare data regulations like HIPAA would make the system suitable for sensitive medical environments.

- **Federated Learning Implementation:** Exploring federated learning approaches would enable model improvement while keeping patient data secure at its source institution.

- **Real-world Validation Studies:** Conducting clinical validation studies in healthcare settings would provide valuable feedback on model performance and user experience in practical scenarios.

- **Multilingual Support:** Adding language localization would make the tool accessible to healthcare providers in non-English speaking regions.

5.2 Conclusion:

The AICTE TechSaksham Medical Diagnosis Project successfully demonstrates the practical application of machine learning in healthcare diagnostics through a user-friendly web application. By implementing multiple disease prediction models within a unified interface, the project addresses a critical need in healthcare for accessible decision support tools that can assist medical professionals in preliminary assessments.

The project's modular architecture enables scalability and extensibility, providing a foundation that can evolve alongside advances in medical machine learning. The implementation of technologies like Flask, scikit-learn, and TensorFlow showcases how modern software development practices can be effectively applied to healthcare challenges.

The system's ability to predict multiple conditions—diabetes, heart disease, hypothyroid, lung cancer, and Parkinson's disease—with reasonable accuracy represents a valuable contribution to the growing field of AI-assisted medical diagnostics. While the tool is designed as a supplement to, rather than a replacement for, professional medical judgment, it offers significant potential for supporting healthcare providers, particularly in resource-constrained settings.

Through this project, we have demonstrated that accessible machine learning tools can be developed to support medical decision-making processes. The open-source nature of the project encourages further community development and adaptation to specific healthcare contexts, potentially improving patient outcomes through earlier detection and intervention.

As healthcare continues to evolve with technological advances, systems like this represent an important bridge between cutting-edge machine learning research and practical clinical applications, contributing to the broader goal of improving healthcare accessibility and effectiveness through thoughtful technology implementation.

REFERENCES

[1] Breiman, L. 2001. Random Forests. Machine Learning. 45, 1 (Oct. 2001), 5–32. DOI:https://doi.org/10.1023/A:1010933404324.

[2] Choudhary, K. 2020. End to End Deployment of Heart Disease Prediction Through Flask With Machine learning Algorithm. The Startup.

[3] Cortes, C. and Vapnik, V. 1995. Support-vector networks. Machine Learning. 20, 3 (Sep. 1995), 273–297. DOI:https://doi.org/10.1007/BF00994018.

[4] Esteva, A., Kuprel, B., Novoa, R.A., Ko, J., Swetter, S.M., Blau, H.M. and Thrun, S. 2017. Dermatologist-level classification of skin cancer with deep neural networks. Nature. 542, 7639 (Feb. 2017), 115–118. DOI:https://doi.org/10.1038/nature21056.

[5] Friedman, J.H. 2001. Greedy Function Approximation: A Gradient Boosting Machine. The Annals of Statistics. 29, 5 (2001), 1189–1232.

[6] Molnar, C. Interpretable Machine Learning.

[7] Rajkomar, A., Dean, J. and Kohane, I. 2019. Machine Learning in Medicine. The New England Journal of Medicine. 380, 14 (Apr. 2019), 1347–1358. DOI:https://doi.org/10.1056/NEJMra1814259.

[8] Wainer, J. 2016. Comparison of 14 different families of classification algorithms on 115 binary datasets. arXiv.